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Multi-scale Dynamic Network for Document Shadow Removal

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Multi-scale Dynamic Network for Document Shadow Removal

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Abstract

Document clarity is pivotal for reliable information transmission, yet shadows remain a pervasive degradation that harms visual quality and downstream processing efficiency. Existing approaches often falter under extreme illumination, struggle with multi-scale shadow patterns, and exhibit limited generalization across document types and capture conditions. We present an end-to-end multi-modal, multi-scale framework that integrates advanced architectures with dynamic feature fusion to adaptively suppress shadows of varying sizes while preserving fine textual details. Concretely, we introduce a Multi-scale Dynamic Network (MDN) that performs scale-aware gating and cross-branch aggregation, enabling the model to emphasize informative cues and attenuate shadow bias at each resolution level. The pipeline is streamlined to reduce manual pre-/post-processing, thereby improving both efficiency and effectiveness in practical document workflows. Experimental results show consistent gains over traditional and deep learning baselines on the Jung and Kligler datasets, confirming superior accuracy and robustness under challenging lighting.

CCS Concepts

• **Computing methodologies → Computational photography**.

Keywords

Document Shadow Removal, Image Shadow Removal, Deep Learning, Image Processing

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1 Introduction

Document shadows degrade legibility, corrupt character strokes, and hinder OCR and layout analysis, which makes their removal essential for digital archives and everyday office workflows. Beyond visual nuisance, shadows act as a multiplicative illumination field that modulates the underlying reflectance, so naive enhancement can erase thin strokes or introduce halos. The problem is compounded by mixed and spatially nonuniform lighting, camera response nonlinearity, and the wide range of shadow scales from glyph-level gaps to page-level casts. A practical solution must disentangle illumination from content, preserve high-frequency textual detail, and adapt its receptive field across scales while remaining memory and latency-conscious.

Classical pipelines based on edge detection, filtering, and morphology [2, 4, 34, 35, 63, 64] provide interpretability but are sensitive to noise and distortion, blur edges, create artifacts, and require manual tuning for each scene. Deep learning methods raise robustness and quality, for example, the document-oriented BEDSR Net [17], the water filling strategy with Lambertian modeling [9], and Laplacian pyramid decoupling in FSENet [13]. However, many existing networks depend on static pyramids or fixed fusion weights, treat shadows primarily as low-frequency components, and rely on a single modality, which limits generalization under extreme illumination. Memory consumption is often high due to multi-level decoders and large feature maps, and edge or text structure is not explicitly protected, leading to over-smoothing of strokes and residual banding. Recent high-resolution datasets and frequency-aware designs improve quality but also underscore the need for dynamic multi-scale and modality-aware fusion and evaluation aligned with readability and OCR rather than image-only metrics [12, 13, 44].

In this paper, we design a Multi-scale Dynamic Network that operates on a feature pyramid and fuses complementary cues in a unified end-to-end pipeline. An RGB branch captures appearance and texture while an auxiliary branch encodes illumination mattes and text or layout priors as additional modalities. When external inputs are unavailable, these priors are derived from the same image and are still treated as a distinct modality. At each pyramid level, a Multi-scale Local Shadow Module performs structure-aware local refinement conditioned on edge and text cues to sharpen boundaries and preserve thin strokes. A Dynamic Scale Fusion Module estimates confidence maps for upward and downward information flow. It performs gated bidirectional aggregation across adjacent resolutions, allowing the network to select the most informative scale without inflating memory. Cross-modal fusion is realized by lightweight gates applied to spatially aligned features from the two branches. Training combines intensity, structural similarity, edge preservation, and illumination smoothness objectives to balance de-shadowing and detail retention while promoting optimization.

In summary, our contributions are as follows: (1) End-to-end document de-shadowing with cross-modal fusion on a feature pyramid, improving efficiency and fidelity in real-world capture. (2) Multi-scale Local Shadow Module for edge-aware shadow localization and stroke preservation, enhancing boundary sharpness across resolutions. (3) Dynamic Scale Fusion Module for confidence-guided multi-scale aggregation with low memory use, adapting to illumination and scale variation.

2 Related work

2.1 Traditional Methods

Traditional methods for document shadow removal typically rely on classical image processing techniques such as image enhancement, filtering, threshold segmentation, and morphological operations [2, 34]. However, these pipelines are often designed for specific lighting assumptions and require careful parameter tuning, which limits adaptability to diverse environments [2, 35]. Many approaches treat shadows as smoothly varying illumination and thus struggle near umbra–penumbra transitions or on textured papers, leading to edge blur, halos, and artifacts [4, 64]. Even document-specific algorithms based on explicit shading estimation (e.g., water-filling with a Lambertian model) can be sensitive to deviations from assumptions and content variations [9]. Overall, robustness and generalization remain limited across lighting, materials, and capture devices, and fine textual details at shadow boundaries are easily lost [63].

2.2 Deep-learning Methods

Deep learning methods [3, 6, 14–16, 18, 19, 21–33, 37–42, 47–50, 50–62, 65, 66, 68–74], particularly CNN and GAN based models, have advanced document shadow removal, yet key challenges remain. They struggle with scene adaptability under extreme lighting and complex, spatially varying shadows, often yielding suboptimal results; for example, FDRNet [67] is sensitive to resolution on the CUHK Shadow dataset. Handling multi-scale shadows is difficult due to variations in size and shape. Generalization across devices and capture conditions is limited, and high computation and memory demands hinder practical deployment. Cross-dataset evaluations frequently show performance drops under different lighting and complex backgrounds. These gaps motivate dynamic, multi-scale, and modality-aware frameworks that prioritize readability, robustness, and efficiency.

3 Method

The overall architecture is an encoder-decoder framework that we have established for document shadow removal. The encoder-decoder backbone network utilizes down-sampling and up-sampling to process the extracted features. The encoder and decoder are connected through skip-connections. Specifically, each encoder or decoder level contains a Multi-scale Dynamic Network (MDN), within which the Multi-scale Local Shadow Module (MLSM) is used to determine the location of the shadows better, and a Dynamic

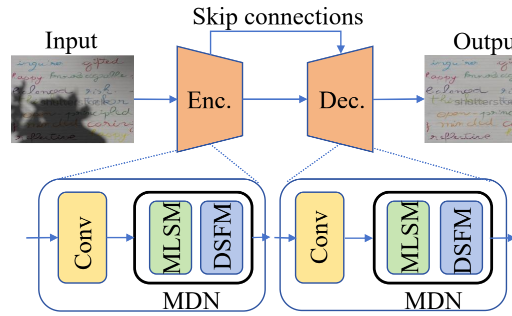

Figure 1: Network architecture. Top: an encoder–decoder backbone with skip connections. Bottom: the encoder and decoder share the same building blocks. MDN denotes the multi-scale dynamic network. MLSM denotes the multi-scale local shadow module, which improves shadow localization across scales. DSFM denotes the dynamic scale fusion module, which aggregates features at different resolutions to enhance multi-scale shadow removal.

Size Fusion Module (DSFM) is used to extract and dynamically fuse multi-scale features.

3.1 Multi-scale Local Shadow Module

This module is designed to detect shadow locations in document images. It operates on shadowed images as input and includes a shadow-localization submodule to identify shadow regions accurately. Given a four-channel feature map, the localization submodule splits it into a three-channel feature map and a single-channel potential map.

Specifically, assuming the input four-channel feature map is F, it can be separated into a three-channel feature map F_c and a single-channel feature map F_s as follows:

$$F_c = \mathrm{Concat}(F_1, F_2, F_3)\,,\ F_s = F_4, \tag{1}$$

where F_1, F_2, F_3, F_4 denote the four channels of the feature map.

The single-channel feature map transforms a sigmoid layer, producing a single-channel shadow region prediction mask. The numerical range of this mask is from 0 to 1, representing the probability of a pixel being part of the shadow region (a higher value indicates a higher likelihood that the pixel belongs to the shadow region). The transformation through a sigmoid layer can be expressed as:

$$S(x) = \frac{1}{1 + e^{-x}}, \tag{2}$$

where x represents the pixel value of the feature map.

At the localization module, we combine the three-channel feature map with the predicted shadow image. After element-wise summation, we get a single-channel predicted shadow image. Finally, we get the predicted shadow image and the actual shadow region image as the input of the DSFM module.

3.2 Dynamic Size Fusion Module

This module is designed for the extraction and dynamic fusion of multi-scale features. It optimizes the prediction accuracy of shadow regions using cross-entropy loss by inputting the predicted shadow

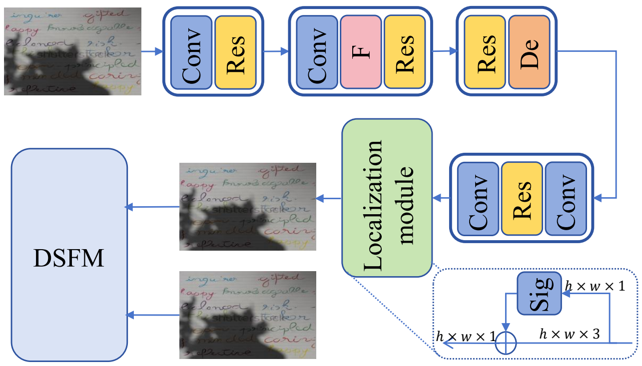

Figure 2: Illustration of the Multi-scale Local Shadow Module (MLSM). *Conv*: convolution; *Res*: residual block; *F*: feature map; *De*: deconvolution (transposed convolution); *c*: concatenation; *sig*: sigmoid; *DSFM*: Dynamic Scale Fusion Module. The localization submodule performs an element-wise summation between the $h\times w\times 3$ feature and the $h\times w\times 1$ potential map, followed by a sigmoid, producing an $h \times w \times 1$ predicted shadow map.

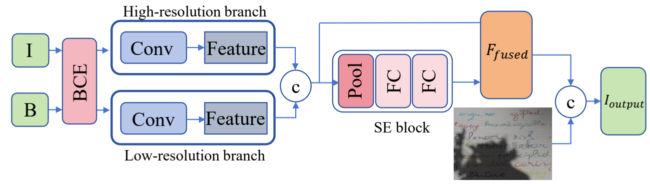

Figure 3: Illustration of the Dynamic Scale Fusion Module (DSFM). The module takes two inputs: B (ground-truth shadow mask) and I (predicted shadow map). A binary cross-entropy (BCE) term measures the discrepancy between B and I. Features are extracted in parallel high-resolution and low-resolution branches to obtain multi-resolution representations. The squeeze-and-excitation (SE) block comprises global average pooling followed by a two-layer fully connected (FC) network. The fused feature F_{fused} is concatenated with B to produce the output I_{out}.

image I and the actual shadow region image B. During the feature extraction phase, the model extracts multi-scale features at both low and high resolutions, fusing them through weighting to capture shadow information across different scales. Additionally, a Squeeze-and-Excitation (SE) block is introduced, which generates attention coefficients through global average pooling and fully connected layers to enhance the representation of important features. Finally, by combining the shadow region probability map and the weighted feature map, a clear de-shadowed image is produced. The loss function consists of shadow region prediction loss and image reconstruction loss. By optimizing these two components, the model can improve performance, successfully remove shadows, and restore clear images.

Firstly, the Binary Cross-Entropy (BCE) loss is used to calculate the discrepancy between the predicted shadow region and the actual shadow region, measuring the difference between the predicted values and the true values at each pixel:

$$L_{shadow} = -\frac{1}{N}\sum_{i=1}^{N}[y_i log(p_i) + (1-y_i)log(1-p_i)], \tag{3}$$

where y_i represents the pixel value of the actual shadow region image, $y_i \in \{0,1\}$; p_i represents the pixel value of the predicted shadow region image, $p_i \in [0,1]$; N is the total number of pixels in the image.

To better capture shadow information at different scales, the network extracts multi-scale feature maps F_{low} and F_{high}, representing low-resolution and high-resolution images, respectively. These feature maps are fused through a weighted sum. The fused feature map is represented as:

$$F_{fused} = \alpha F_{low} + \beta F_{high}, \tag{4}$$

Where α and β are learnable weights that control the contribution of features at different scales.

Next, we employ the Squeeze-and-Excitation (SE) block, which adaptively adjusts the weights of each channel in the feature map based on global information. The specific steps include: performing global average pooling on the input feature map to obtain the global information for each channel:

$$z_c = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W} F_c(i,j), \tag{5}$$

Where z_c represents the global description for the c-th channel, where H and W are the height and width of the feature map, respectively, and $F_c(i,j)$ denotes the value of the c-th channel at position (i, j) in the feature map F.

The global description for each channel is then processed through a two-layer fully connected network (FC) to obtain the attention coefficients for each channel:

$$s_c = \sigma(W_2 \cdot \delta(W_1 \cdot z_c)), \tag{6}$$

Where W_1 and W_2 are the weight matrix of the fully connected layer, δ is the RELU activation function, σ is the sigmoid activation function, s_c is the attention coefficient of the channel.

Finally, the weighted feature map is obtained by applying the attention coefficients to the original feature map:

$$F_{attended} = F_{fused} \cdot s. \tag{7}$$

Next, we fuse the predicted shadow region map and the real shadow region map with the feature maps output by the network to remove the shadow and reconstruct the clear image. Assuming we obtain a shadow probability map p_{shadow} through the network, which is used to indicate the shadow probability at each pixel. The final de-shadowed image can be represented as I_{output}:

$$I_{output} = (1 - p_{shadow}) \cdot B + p_{shadow} \cdot F_{attended}, \tag{8}$$

Where p_{shadow} is the predicted shadow region probability map, $F_{attended}$ is a multi-scale feature map weighted by an attention mechanism.

To train the network, we need to define a total loss function, which consists of two parts. The total loss includes a shadow region loss:

$$L_{shadow} = -\frac{1}{N}\sum_{i=1}^{N}[y_i log(p_i) + (1-y_i)log(1-p_i)], \quad (9)$$

an image reconstruction loss:

$$L_{reconstruction} = \frac{1}{N}\sum_{i=1}^{N}(I_{output}(i) - I_{groundtruth}(i))^2, \quad (10)$$

Where $I_{groundtruth}(i)$ represents the actual shadow region image loss.

With the final total loss being the weighted sum of these two losses:

$$L_{total} = L_{shadow} + L_{reconstruction}, \quad (11)$$

where λ_1 and λ_2 are weight hyperparameters used to balance shadow region prediction and image reconstruction losses.

4 Experiments

4.1 Experiment Setup

4.1.1 Datasets and Metrics. To evaluate our method, we conduct experiments on two public document-shadow datasets. The Jung dataset [9] is a standard benchmark with approximately 67 training images and 20 validation images, spanning diverse page layouts (single-page, double-page, partial-page) and a wide range of shadow conditions, including natural, artificial, and reflective shadows. The Kligler dataset [10] contains about 272 training images and 28 validation images, with shadows cast from varied light directions (natural and artificially projected) and backgrounds ranging from uniform colors to complex textures. For evaluation, we report Root Mean Square Error (RMSE), Peak Signal-to-Noise Ratio (PSNR, in dB), and Structural Similarity (SSIM).

4.1.2 Implementation Details. We compare the proposed method with representative document shadow removal approaches, including classical pipelines (edge detection, filtering, morphology) and superior deep models: Wang [46], Wang [43], Shah [36], Kligler [10], BEDSR-Net [17], DHAN [1], LG-ShadowNet [20], DC-ShadowNet [8], Mask-ShadowGAN [7], SP+M-Net [11], ST-CGAN [45], and ShadowFormer [5]. For learning-based baselines, we follow official implementations and default hyperparameters when available. Note that the comparison methods' results are obtained from the published papers. Therefore, we do not show the comparison figure of our method with existing methods.

We implement our model using PyTorch and optimize with Adam. The initial learning rate is set to 1×10^{-4}. Training is performed on an NVIDIA Tesla T4 with a batch size of 4 for 600 epochs.

4.2 Comparison with State-of-the-art Methods

As shown in Table 1, we compare the proposed MDN with both classical and learning-based approaches across multiple input resolutions, and our method achieves the best performance on most metrics. For methods without publicly available code, we re-implement the pipelines following the original papers and default settings to ensure a fair comparison.

Several classical techniques depend on physics-based assumptions that are often violated in real scenes, which yields unstable performance across datasets. Conversely, some deep models exhibit limited robustness and generalization, with notable variation under differing illumination, backgrounds, and capture conditions.

Table 1: Quantitative comparison with state-of-the-art methods on the Jung and Kligler datasets.

Method	Jung			Kligler		
	PSNR	SSIM	RMSE	PSNR	SSIM	RMSE
Input	13.01	0.82	60.85	13.26	0.80	56.73
Wang et al. [46]	11.17	0.78	73.27	15.73	0.82	44.04
Wang et al. [43]	9.11	0.71	90.99	15.38	0.72	48.03
Shah et al. [36]	14.69	0.80	47.97	8.36	0.70	97.86
Kligler [10]	18.56	0.81	31.18	18.15	0.81	32.40
BEDSR-Net [17]	21.51	**0.85**	22.58	22.31	0.75	20.86
DHAN [1]	20.58	0.82	25.95	25.66	0.84	15.49
LG-ShadowNet [20]	19.99	0.84	27.69	26.29	0.87	14.38
DC-ShadowNet [8]	21.06	0.83	24.32	26.88	0.90	13.76
Mask-ShadowGAN [7]	19.41	0.82	29.16	25.79	0.87	15.01
SP+M Net [11]	20.04	0.84	29.13	18.85	0.86	29.57
ST-CGAN [45]	13.93	0.33	52.51	12.17	0.44	63.22
ShadowFormer [5]	20.61	**0.85**	26.96	17.26	0.77	36.31
Ours	**22.33**	**0.85**	**20.33**	**27.21**	**0.92**	**13.55**

Table 2: Ablation Study on Jung's dataset.

Method	PSNR↑	SSIM↑	RMSE↓
Input	12.93	0.79	61.34
Ours w/o MLSM	21.37	0.84	22.25
Ours w/o DSFM	21.73	0.84	21.43
Ours w/o Localization Module	21.86	0.84	21.06
Ours Full	**22.33**	**0.85**	**20.33**

In images containing multi-scale shadows, many deep networks are not sufficiently scale-adaptive, which leads to incomplete removal or text over-smoothing; classical methods also struggle in these cases. By contrast, MDN's scale-aware localization and confidence-guided fusion provide consistent gains on mixed-scale shadows.

Figure 4 shows clear improvements on the Jung dataset. Our method removes shadows at both page-scale and glyph-scale, yielding an even background. Halo artifacts at umbra-to-penumbra transitions are largely suppressed. Text strokes remain sharp, local contrast is restored, and small graphical details are preserved without excessive smoothing. Color and luminance stay consistent across layouts, and overall readability is clearly improved.

4.3 Ablation Study

We conducted an ablation study on the Jung dataset to examine the effectiveness of several components in our algorithm, which is shown in Table 2. These components include the MLSM, DSFM, and the localization module within MLSM. We first evaluated the proposed MLSM module. This module is primarily responsible for locating the position of the shadows and has made a significant contribution to the entire network. Additionally, we assessed the

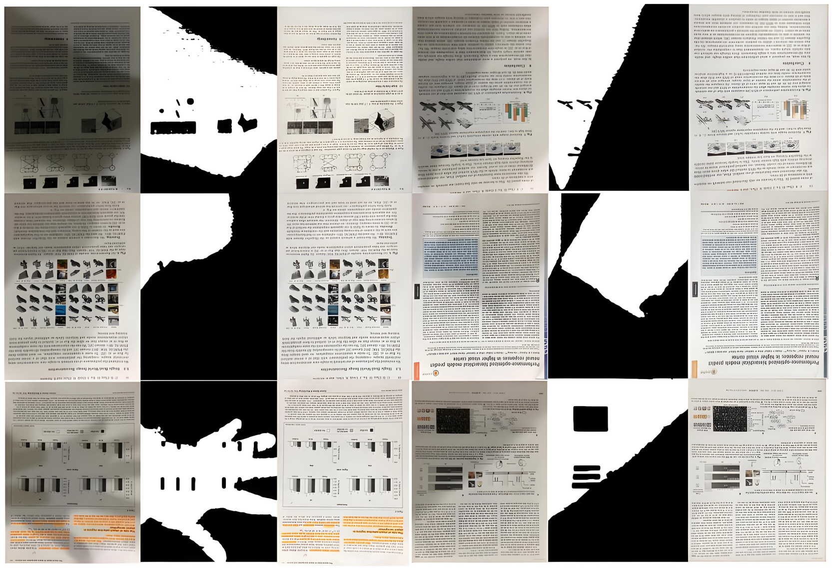

Figure 4: Illustration of the results on Jung dataset. Different color rectangles represent different sizes of shadows.

DSFM module, which is mainly responsible for dynamic fusion and also contributes to the network's performance. Although each module individually contributes to improving the shadow removal performance, their combined network produced the best results. Furthermore, we also evaluated the localization module within MLSM. We found that the localization module can better determine the position of the shadows, and its removal had a substantial impact on the network.

5 Conclusion

We present an end-to-end Multi-scale Dynamic Network (MDN) for document shadow removal. By integrating the Multi-scale Local Shadow Module (MLSM) and the Dynamic Scale Fusion Module (DSFM), the model adaptively localizes shadows and fuses information across resolutions. On the Jung and Kligler datasets, MDN surpasses classical and learning-based baselines, achieving higher PSNR and SSIM and lower RMSE, while preserving text strokes and background fidelity.

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